Soroosh Noorzad

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Research Interests

Machine Learning for Electronic Design Automation (EDA), with a focus on graph neural networks, RF circuit modeling, surrogate modeling, and generative inverse design.

Education

University of Utah — Salt Lake City, Utah, US
Ph.D. in Electrical and Computer Engineering — *2025–Expected 2030*
Advisor: Morteza Fayazi

Sharif University of Technology — Tehran, Iran
M.Sc. in Electrical Engineering, Digital Systems — *2020–2023*
Advisor: Khosrow Haj Sadeghi

Qom University of Technology — Qom, Iran
B.Sc. in Electrical Engineering, Telecommunication — *2010–2014*

Research Experience

- **FInDeR: ML-Based RF Circuit Inverse Design** — University of Utah & Texas A&M University
  Supervisors: Milad Koohi, Morteza Fayazi — *2025–Present*
  - Developed a multihead CVAE framework for inverse design of passive RF circuit topologies and component parameters from target frequency responses.
  - Integrated a frequency-conditioned GNN and CNN residual refiner for surrogate-based candidate ranking and circuit-model verification.
  - Achieved 1.037 dB mean verified MAE across 100 held-out targets, with 3.79× faster evaluation and 32.1% lower MAE than a genetic algorithm baseline.
- **TARGet: Topology-Aware RF Circuit Modeling** — University of Utah
  Supervisor: Morteza Fayazi — *2025–2026*
  - Developed a topology-aware RF circuit modeling framework combining GNNs, sub-circuit connectivity-aware networks, and S-parameters.
  - Achieved sub-1% error with up to 35.5× less training data and 9.7× higher accuracy under a 1% error threshold.
  - Demonstrated zero-shot generalization to unseen sub-circuit topologies, reducing NMAE by up to 45.1%.
- **ML for Multi-access Edge Computing-based networks** — Sharif University of Technology
  Supervisor: Khosrow Haj Sadeghi — *2021 - 2023*
  - Implemented an ML-based computation offloading system using a near-optimal decision-making approach.

Publications

TARGet: Topology-Aware Fusion-based Radio Frequency Circuit Functional Modeling using Graph Neural Networks
Soroosh Noorzad, Sebastian Bodero, Morteza Fayazi
IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2026.

SINA: A Fully Automated Circuit Schematic Image to Netlist Generator Using Artificial Intelligence
Saoud Aldowaish, Yashwanth Karumanchi, Chen Chiang, **Soroosh Noorzad**, Morteza Fayazi
Design, Automation and Test in Europe Conference and Exhibition (DATE), 2026.

MuaLLM: A Multimodal Large Language Model Agent for Circuit Design Assistance with Hybrid Contextual Retrieval-Augmented Generation
Pravallika Abbineni, Saoud Aldowaish, Colin Liechty, **Soroosh Noorzad**, Ali Ghazizadeh, Morteza Fayazi
IEEE Asia and South Pacific Design Automation Conference (ASP-DAC), 2026.

Manuscripts Under Review

FInDeR: Fast Generative Inverse Design of Passive RF circuits via Surrogate-Based Candidate Selection
Soroosh Noorzad, Axton Grams, Abdullah Hammami, Milad Koohi, Morteza Fayazi
Submitted to Design, Automation and Test in Europe (DATE), 2027.

Technical Skills

- **Programming Languages:** Python, C/C++, PHP, MATLAB, SQL
- **Frameworks & Libraries:** PyTorch, TensorFlow 1&2, Scikit-learn, OpenCV, PyQt5
- **Tools & Technologies:** Linux, CUDA, Bash, LaTeX, Git, Anaconda, VS Code, Microsoft Office

Selected Projects

- **Laser-Guided Remote Object Targeting System** 2025
  - Developed a real-time object detection and laser-targeting system integrating ESP32-CAM, STM32, wireless communication, computer vision, and servo-based control.
- **CNN-Based Aluminum Welding Defect Classification** 2024
  - Improved defect classification accuracy by 4% over a published baseline using PyTorch and Scikit-learn.
- **GPU and CPU Parallel Computing** 2022
  - Implemented CUDA-based matrix multiplication and parallel prefix-sum (Blelloch scan) algorithms, along with CPU-based parallel merge using POSIX threads in C.
- **Joint Object Detection and Depth Estimation** 2021
  - Integrated YOLOv3-based object detection and U-Net-based depth estimation (trained on NYU Depth V2) into a PyQt graphical application.
- **Sentiment Analysis Using Deep Learning** 2021
  - Trained an LSTM-based sentiment classifier on the Sentiment140 dataset using TensorFlow 2 and GloVe embeddings.
- **Remote Camera Monitoring and Face Detection** 2021
  - Developed a remote camera system integrating C/C++ networking (Boost.Beast), OpenCV, PHP, and MariaDB for image processing, storage, and remote access.
- **MQTT Protocol and Linux Services** 2021
  - Implemented Linux services for MQTT-based communication using C/C++, Make, and CMake.

Teaching Experience

- **Teaching Assistant for "ECE/CS 3700: Fundamentals of Digital System Design"** University of Utah
  Dr. Morteza Fayazi *Fall 2026*
  - Design and grade homework assignments and hold student office hours.
- **Head Teaching Assistant for "Real-time Embedded Systems"** Sharif University of Technology
  Dr. Iman Gholampour *Spring 2022 & 2023*
  - Led a team of teaching assistants in developing and grading assignments and projects.
  - Mentored student projects involving real-time scheduling, embedded HTTP servers, and RESTful APIs using Python and C/C++.
  - Delivered tutorials on web services, client-side scripting, and web security.
- **Teaching Assistant for "Data Telecommunication Networks"** Sharif University of Technology
  Dr. Mohammad Reza Pakravan *Fall 2021*
  - Designed and graded assignments on physical-layer concepts and MATLAB-based data-link-layer simulations.

Honors and Awards

- **Awarded - 2026**: Student Scholarship Program Award for International Conference on Computer-Aided Design (ICCAD).
- **Awarded - 2026**: Graduate Student Professional Development Supplement Award, University of Utah, Graduate School.
- **Ranked** 3rd **- 2026**: Poster competition at the University of Utah AI Convergence Conference, among 74 presentations.
- **Ranked** 23rd **- 2020**: Iranian national M.Sc. entrance exam (Konkoor) for Electrical Engineering, among 40,000+ test-takers.

Professional & University Service Activities

Reviewer for:

- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)

Languages

Persian: Native
English: Full Professional Proficiency